ECG Arrhythmia Beat Classification Using a Convolutional Bidirectional LSTM Deployed as a Tool for a Retrieval-Augmented LLM Agent

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Abstract / Problem Statement

Cardiac arrhythmias are usually identified from long ambulatory electrocardiogram (ECG) recordings, where manual beat-by-beat review is slow and error-prone. Many published deep-learning heartbeat classifiers report near-perfect accuracy, but they often split beats randomly, so beats from the same patient appear in both training and test data and the reported performance does not carry over to new patients. This project builds a heartbeat classifier that is evaluated on unseen patients and makes it usable through a conversational assistant.

Single-lead recordings from the MIT-BIH Arrhythmia Database are band-pass filtered, segmented around each R-peak and grouped into the AAMI classes normal (N), supraventricular ectopic (S) and ventricular ectopic (V). A compact convolutional bidirectional long short-term memory (Conv-BiLSTM) network of about 50,000 parameters combines beat morphology with rhythm features expressed relative to each patient's own heart rate. Model selection uses patients held out from training, and final testing uses 22 patients never seen during development, reported with patient-level bootstrap confidence intervals and compared against baselines.

On the unseen patients the model reaches 94.5% accuracy and a macro-F1 score of 0.65, against 0.53 for a rhythm-only baseline. It detects 80% of ventricular beats with 85% precision, and its probabilities are well calibrated. Supraventricular beats remain difficult, and this limitation is reported explicitly.

The model is deployed in a FastAPI web service with an ECG signal-quality check. An OpenAI language model acts as the agentic core: it decides when to call the classifier as a tool and when to retrieve reference documents from a persistent Chroma vector database (retrieval-augmented generation), with a Gemini model as automatic fallback. Answers about an uploaded ECG therefore rest on both the measurement and cited sources. The system is verified by 119 automated tests, including live tests against the real language-model APIs.

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